full lighting system that can be remotely controlled and would respond to human presence, or an automated home theatre comprising a smart TV with smart ambient lighting.

Any smart home automation system today is generally a central hub that can be configured to control a bunch of smart devices, sensors and switches, all of which communicate with the hub using certain communication protocols. The hub, in turn, is instructed through an app or the web. The main takeaway is the distribution of monitoring and computing functions between the hub and the remote app. For example: in a smart lighting system, a hub would act as the central interface between multiple smart devices, say, a bulb and a door contact sensor. The smart devices and hubs communicate using certain common communication technologies, and an app would be used to control the lighting system.

If you're still unclear about the role of the hub, you can draw close parallels between it and a standard Wi-Fi router. In simple terms, both are devices that route signals from multiple sources to one another. In a few products, the hub and router are integrated together, thus reducing the need for two devices. However, in the cases when they are separate, the hub, which needs to be Internet enabled to function, is connected to the router. So basically, a smart hub provides a centralized method to control all your smart devices, as they can connect all your devices to the cloud and consolidate all apps into the one provided by the hub manufacturer.

Hubs/Controllers

When we talk of smart devices, it encompasses an almost never ending field – possibilities include lighting systems, electrical appliances like irons, microwaves, geysers, lamps, security systems, multiple wall outlets and switches, smoke detectors and a lot more. Many of these devices are used in conjunction with sensors, to allow for greater versatility in their control and instruction. Since having multiple apps to control all these devices is cumbersome, today's home automations systems use hubs, also called controllers, to provide a place of convergence for all data, and to synchronize its flow too. Hubs, in general, are defined as network devices where data is forwarded in through multiple devices, and then sent out to the appropriate device. Since these hubs work on the same principle as routers, as mentioned earlier, they have inbuilt switches, which are the components responsible for the forwarding of data to the right device. Examples of hubs and controllers in the market are Samsung's SmartThings, Insteon